

Resource-Aware Amplitude Amplification for Hybrid Quantum–Classical Workflows: A Reproducible Max-Cut Benchmark with QAOA and Classical References

Ercan Erkalkan^{a,*}

^aMarmara University, Vocational School of Technical Sciences, Department of Electronics and Automation, Artificial Intelligence Operator Program, Mehmet Genç Campus, Istanbul, 34865, Turkey

Abstract

Amplitude amplification is usually assessed through query complexity, but hybrid workflows also incur target construction, oracle realization, repeated circuit depth, fixed per-trial work, and classical optimization. We present a reproducible weighted Max-Cut benchmark that separates these effects without hardware-speedup claims. Operational thresholds use graph-observable quantities—the uniform-random expected cut and a Laplacian spectral upper bound—without the unknown optimum C^* . Exact enumeration is reserved for retrospective validation on an 18-instance primary tier, a 63-instance coverage tier, and a 45-instance connected, non-bipartite tier through $n = 14$. A reversible accumulator/comparator model couples threshold-oracle cost to logical depth. With zero fixed per-trial overhead, all 255 primary and 360 coverage conditions select no amplification under the declared probability-per-resource objective; median zero-attenuation break-even overheads are 0.163 and 0.168 oracle-cost units. A BBHT unknown-solution-count layer removes the marked fraction from the executed growth schedule and quantifies its information cost. QAOA at $p = 1, 2, 3$ uses instance-conditioned initialization; a nested 20-start audit (1,080 L-BFGS-B runs) preserves all three positive depth contrasts after Holm correction, while no COBYLA-minus-L-BFGS-B contrast survives correction under an equal 128-evaluation cap. A numerically certified Goemans–Williamson SDP closes all 63 primary/connected-validation relaxations within 10^{-6} , and seeded rounding reaches an exact optimum on 62/63 instances. Simulated annealing and tabu search solve every primary micro-instance at least once even at low evaluation budgets. A compiler-locked Qiskit 2.4.2 layer adds executable transpilation evidence while keeping physical-hardware claims explicitly out of scope.

Keywords: quantum computing, amplitude amplification, QAOA, Max-Cut, hybrid quantum–classical systems, reproducible benchmarking

1. Introduction

Quantum algorithms are often first compared through asymptotic query complexity. Grover search and its generalization through amplitude amplification establish a quadratic reduction in the number of oracle uses under the corresponding black-box assumptions [1, 2]. That abstraction is fundamental, but it does not by itself answer a systems question: when a search procedure is embedded in a finite hybrid workflow, which configuration should actually be preferred under explicit resource assumptions?

The distinction matters because a practical workflow has more structure than an oracle count. State preparation, oracle construction, diffusion, repeated circuit execution, classical control, verification, compilation, and optimization can all contribute to the useful-work budget. The importance of quantum–classical integration is increasingly visible in the systems literature, including work on tightly coupled quantum–classical execution and quantum/HPC integration [3, 4]. At the same time, classical simulation remains essential for validating small and

intermediate quantum workloads, while its own scaling limitations must be made explicit [5]. These developments motivate benchmark designs that separate evidence classes instead of collapsing analytical, simulated, compiled, and hardware observations into one performance label.

Variational algorithms create an additional difficulty. QAOA is a hybrid algorithm in which circuit depth and classical parameter optimization interact [6]. The quality obtained from a given depth can depend strongly on the optimizer and its initialization [7, 8, 9]. Reporting a single favorable run can therefore overstate typical behavior. A neutral comparison should expose this variability and should include a classical reference, especially when the tested instances are small enough that classical methods remain highly competitive.

This paper addresses these issues through a compact but deliberately auditable Max-Cut benchmark. The research question is:

When a Max-Cut target threshold is constructed from graph-observable quantities rather than the unknown optimum, how does amplitude-amplification configuration respond to target strictness and resource assumptions, and how do amplitude amplification, QAOA, uniform sam-

*Corresponding author.

Email address: ercan.erkalkan@marmara.edu.tr (Ercan Erkalkan)

pling, and classical local search behave under the same target-hit event $C(x) \geq \tau$?

The goal is not to establish quantum advantage. Instead, the study contributes four elements. First, it separates *threshold construction* from *threshold evaluation*: the operational threshold uses only the graph weights and Laplacian spectrum, whereas C^* is used only retrospectively in the exact tier to characterize feasibility and realized quality. Second, it evaluates threshold-aware amplification and QAOA over the complete primary exact suite of 18 weighted Max-Cut instances, audits connectivity and bipartiteness explicitly, stress-tests seed and topology dependence in a separate 63-instance exact coverage tier, and adds a 45-instance connected, non-bipartite exact validation tier through $n = 14$; it also retains a separate C^* -normalized calibration axis, quantifies fixed-depth regret under multiplicative marked-fraction error, and evaluates a BBHT schedule that does not consume the marked fraction. Third, it introduces a common target-hit quantity $P_\tau = \Pr[C(x) \geq \tau]$ for amplitude amplification, QAOA, uniform sampling, and classical local search while keeping method-specific resource accounting separate. Fourth, it binds all results to one machine-readable workflow and preserves explicit evidence boundaries between exact, simulated, analytical, compiler, and hardware claims.

The evidence set includes analytical amplitude amplification calculations, exact classical enumeration, ideal-statevector QAOA simulation, synthetic resource sensitivity, host-side scaling diagnostics, reproducible classical heuristics, and a compiler-locked Qiskit transpilation layer on a declared synthetic topology. No physical-backend execution, device calibration, queue-time, or empirical hardware timing result is reported. This boundary shapes the language used throughout the paper.

2. Related work and positioning

2.1. Amplitude amplification and stopping depth

Grover’s search algorithm demonstrates quadratic improvement in black-box search under ideal oracle assumptions [1]. Brassard et al. generalized the mechanism as amplitude amplification, where a procedure that prepares a good state with probability a can be amplified using repeated reflections [2]. Because standard Grover rotations are oscillatory, stopping depth is part of the algorithmic design rather than a monotone “more is better” parameter. Fixed-point variants address uncertainty in the marked fraction through modified phase schedules [10]; the present paper does not implement a fixed-point search, but the broader literature reinforces the need to state which amplitude-amplification regime is being benchmarked.

The present work focuses on standard amplitude amplification with a threshold-defined marked set. Exact enumeration in the small benchmark tier is used to measure the fraction of states satisfying the declared numeric threshold, not to hand the oracle a pre-enumerated optimum set. The novelty is not a new quantum-search theorem; the number of amplification rounds

is treated as a systems configuration selected under an explicit normalized resource model.

The information requirement associated with the marked fraction is distinct from threshold construction. Boyer et al. showed that quantum search can be organized when the number of solutions is not known in advance, providing the BBHT reference point for unknown-solution-count search [11]. The benchmark therefore uses two complementary controls: multiplicative marked-fraction misspecification for an oracle-informed fixed-depth selector, and a BBHT schedule whose growth rule does not consume ρ_τ . Exact Tier-I ρ_τ is used only retrospectively to evaluate the expected BBHT success/resource cost. Quantum counting and fixed-point search remain outside the implemented evidence set.

2.2. QAOA depth and optimizer variability

QAOA alternates problem and mixer unitaries and uses a classical optimizer to set variational parameters [6]; recent surveys emphasize that performance claims depend on depth, parameterization, optimizer behavior, instance structure, sampling, and the classical reference against which QAOA is judged [12]. Optimizer behavior can materially affect the final result; comparative and learning-based studies report substantial dependence on optimization strategy and computational budget [7, 13]. Parameter initialization can also cause convergence to poor local minima, motivating annealing-inspired schemes and explicit analysis of instance-dependent initialization [8, 14]. Depth itself is a consequential hyperparameter, and automatic depth-selection methods have been proposed to reduce unnecessary circuit complexity [9]. Sampling-frequency analyses further show that claims of practical QAOA advantage require explicit accounting of the number of circuit samples needed to reach a target solution quality [15].

These findings motivate the protocol used here: each depth is optimized independently from multiple random initializations, and the reported central statistic is a median across runs. The benchmark therefore does not assume that a larger depth must be better, nor does it identify the best single run with typical performance.

2.3. Quantum systems context

Recent Future Generation Computer Systems work on quantum computing emphasizes hybrid systems, practical algorithm design, simulation, resource estimation, and infrastructure [4]. Hybrid scientific-workflow studies formalize the mapping and management of quantum and classical execution stages [16], while QC/HPC integration frameworks place quantum resources inside broader scientific-computing ecosystems [17]. QHDL addresses tight quantum–classical interaction in the control stack [3], simulator benchmarking studies the computational consequences of circuit structure for statevector and tensor-network execution [5], and large-scale resource-estimation work in FGCS illustrates why logical algorithms must not be conflated with physical implementation cost [18]. Applied QAOA work in the same journal also illustrates the role of hybrid problem formulation and initialization in systems-oriented optimization workloads [19].

Quantum benchmarking also spans several abstraction levels, and recent benchmarking guidance stresses that a benchmark must state which layer of a quantum computer or workflow it actually measures [20]. QASMBench provides low-level OpenQASM workloads for evaluating devices, compilers, schedulers, and simulators [21]; SupermarQ emphasizes scalable, hardware-agnostic application-level workloads and coverage-oriented suite construction [22]; and application-oriented benchmarking frameworks evaluate end-user-relevant quality and execution-pipeline behavior across representative algorithms [23, 24]. The present study is deliberately narrower: it does not propose a general-purpose hardware benchmark suite. Instead, it contributes a problem-specific, exact-ground-truth microbenchmark for separating target construction, algorithm configuration, logical resource accounting, optimizer variability, and evidence class before hardware-level claims are attempted.

The benchmark sits at the algorithm–resource interface. It remains pre-hardware, but now includes one compiler-locked layer while keeping its analytical resource template, compiler realization, and any future physical-backend evidence as distinct claim classes.

3. Evidence model and benchmark design

3.1. Evidence classes

Every result in this study belongs to one of five evidence classes:

1. analytical standard amplitude-amplification and BBHT unknown-solution-count calculations;
2. exact classical enumeration of small Max-Cut instances;
3. ideal-statevector QAOA simulation, including paired optimizer-budget robustness;
4. synthetic attenuation and normalized resource-cost sensitivity;
5. classical references: a numerically certified Goemans–Williamson SDP relaxation with seeded hyperplane rounding, a simple multistart hill climber, and fixed-objective-evaluation-budget simulated annealing and tabu search.

The terms *hardware*, *measured post-compilation depth*, *device calibration*, and *wall-clock quantum speedup* are reserved for evidence not present in this paper. Optimizer execution time is logged for reproducibility but is not interpreted as a portable performance metric because it depends on the host environment and implementation.

3.2. Exact weighted Max-Cut suite

For a weighted undirected graph $G = (V, E)$ with binary assignment $x_i \in \{0, 1\}$, the cut value is

$$C(x) = \sum_{(i,j,w) \in E} w(x_i \oplus x_j). \quad (1)$$

Table 1: Structure of the exact benchmark suite. Each n contains three density targets and two graph seeds.

n	instances	edge range	states	median ρ_{opt}
8	6	7–21	256	0.01172
10	6	11–34	1024	0.00391
12	6	16–50	4096	0.00049

The benchmark uses $n \in \{8, 10, 12\}$ vertices, target edge densities $\{0.25, 0.50, 0.75\}$, graph seeds $\{17, 42\}$, and integer weights sampled uniformly from $\{1, \dots, 9\}$. Each $(n, \text{density}, \text{seed})$ configuration initializes its own versioned deterministic random stream, so density levels sharing the same external seed are reproducible but are not prefixes of one shuffled edge list. Exactly $m = \text{round}(\text{density} \binom{n}{2})$ edges are selected without replacement. The random component is therefore a fixed-edge-count/fixed-density random simple-graph design rather than an Erdős–Rényi $G(n, p)$ draw. This gives 18 deterministic instances. Since the largest state space has only $2^{12} = 4096$ assignments, the complete search space is enumerated to obtain the exact optimum C^* and the set

$$\mathcal{M} = \{x : C(x) = C^*\}. \quad (2)$$

The exact optimum-state fraction $\rho_{\text{opt}} = |\mathcal{M}|/2^n$ is retained as ground truth. For retrospective calibration only, the exact tier also defines

$$\tau_q^{\text{cal}} = \lceil qC^* \rceil, \quad \mathcal{M}_{\tau_q}^{\text{cal}} = \{x : C(x) \geq \tau_q^{\text{cal}}\}, \quad \rho_{\tau_q}^{\text{cal}} = \frac{|\mathcal{M}_{\tau_q}^{\text{cal}}|}{2^n}, \quad (3)$$

for $q \in \{0.90, 0.95, 1.00\}$. These C^* -normalized thresholds are not presented as operational target construction; they provide a controlled quality axis for exact-tier sensitivity and consistency checks.

3.3. Expanded seed and topology coverage

The primary 18-instance exact suite is retained unchanged so that the QAOA protocol and the instance-clustered statistical analysis remain traceable to their original fixed benchmark. To test whether threshold and architecture-informed conclusions are artifacts of only two random seeds, a separate expanded exact coverage tier is added. The random-weighted component uses the same $n \in \{8, 10, 12\}$ and density targets $\{0.25, 0.50, 0.75\}$ with graph seeds $\{17, 42, 73, 101, 211\}$, yielding 45 exact random instances; the original 18 are an exact subset of these 45.

Topology coverage adds three deterministic simple-graph families at each n , with independent integer edge weights generated from seeds 17 and 42. The *cycle* family contains the n ring edges. The *3-regular circulant* family augments the cycle with an opposite-vertex perfect matching. The *two-community* family uses two equal-size cliques connected by two bridges. This gives $3 \times 3 \times 2 = 18$ structured weighted instances and $45 + 18 = 63$ exact coverage instances overall. Each coverage

graph is exhaustively enumerated because $n \leq 12$. The coverage tier records exact optima, operational-threshold feasibility, a 64-start hill-climbing sanity diagnostic, and the architecture-coupled selector at $\varepsilon_D \in \{0, 10^{-3}\}$. It does not enlarge the QAOA hypothesis-testing sample beyond the 18 primary instances; the additional coverage instances remain descriptive stress tests.

3.4. Structural graph audit and connected validation

The primary and expanded coverage suites were originally fixed by deterministic edge-count sampling rather than by a connectivity constraint. Because disconnected components and bipartite structure can increase optimum-state degeneracy or make Max-Cut structurally easier, every exact graph is now accompanied by an explicit structural ledger containing component count, isolated-vertex count, bipartiteness, cycle rank, total edge weight, and exact optimum degeneracy. These descriptors are diagnostic metadata; they do not enter threshold construction or the QAOA objective.

The original 18 primary instances are deliberately retained unchanged to avoid post hoc replacement of unfavorable or structurally simple cases. Instead, a separate validation tier conditions the same fixed-density sampling design on two graph-only properties: connectedness and non-bipartiteness. The tier uses $n \in \{10, 12, 14\}$, density targets $\{0.25, 0.50, 0.75\}$, and seeds $\{17, 42, 73, 101, 211\}$, producing 45 deterministic instances. For each $(n, \text{density}, \text{seed})$ tuple, a versioned rejection stream repeatedly samples exactly $m = \text{round}(\text{density} \binom{n}{2})$ edges until the graph is connected and non-bipartite; edge weights are then drawn uniformly from $\{1, \dots, 9\}$. Exact enumeration remains feasible through $2^{14} = 16,384$ states. Conditioning on connectedness and non-bipartiteness is not presented as a proof of computational hardness; it removes two identifiable structural confounders while preserving an exact-ground-truth validation layer.

3.5. Operational threshold construction without C^*

Let $W = \sum_{e \in E} w_e$. Under uniformly random independent cut bits, the expected weighted cut is

$$\mu_{\text{rnd}} = \frac{W}{2}. \quad (4)$$

Let L be the weighted graph Laplacian. Since $C(x) = \frac{1}{4} s^T L s$ for a spin encoding $s \in \{-1, +1\}^n$, a graph-derived upper bound is

$$U_{\text{spec}} = \min \left\{ W, \frac{n \lambda_{\max}(L)}{4} \right\}. \quad (5)$$

The operational threshold is then

$$\tau_{\lambda}^{\text{op}} = \left[\mu_{\text{rnd}} + \lambda (U_{\text{spec}} - \mu_{\text{rnd}}) \right], \quad \lambda \in \{0.25, 0.40, 0.55\}. \quad (6)$$

No C^* value or optimum-state identity is used in Eq. (6). In Tier I, exhaustive enumeration is performed only after the threshold has been fixed, yielding an exact validation label for feasibility, ρ_{τ} , and the retrospective ratio $\tau_{\lambda}^{\text{op}}/C^*$. An operational target

Table 2: Operational thresholds constructed without C^* . Feasibility, τ/C^* , and ρ_{τ} are retrospective exact-tier validation quantities and are not inputs to threshold construction.

λ	feasible	median τ/C^*	range τ/C^*	median ρ_{τ}	construction
0.25	18/18	0.790	0.625–0.878	0.1816	no C^*
0.40	18/18	0.865	0.700–0.983	0.0566	no C^*
0.55	15/18	0.933	0.775–1.000	0.0156	no C^*

may therefore be declared infeasible by exact validation; the construction does not silently lower it to the optimum.

The three declared operating points are embedded in a separate dense audit with $\lambda = 0, 0.05, \dots, 1.00$. This 21-point sweep applies exactly the same C^* -independent construction and records feasibility, retrospective τ/C^* , ρ_{τ} , and the zero-fixed-overhead logical selector only after each threshold is fixed. The dense sweep is not used to choose the three declared levels after observing performance; it is a sensitivity control for whether the reported points sit on a broader strictness trend.

3.6. Stopping-depth robustness to marked-fraction error

Operational threshold construction does not imply that the corresponding marked fraction ρ_{τ} is known at deployment time. To isolate this second information requirement, the benchmark perturbs the exact Tier-I validation fraction multiplicatively,

$$\widehat{\rho}_{\tau} = f \rho_{\tau}, \quad f \in \{0.50, 0.75, 0.90, 1.00, 1.10, 1.25, 1.50\}, \quad (7)$$

with clipping to $(0, 1]$. The selector chooses $\widehat{k}$ using $\widehat{\rho}_{\tau}$, but the selected configuration is then evaluated under the true exact-tier ρ_{τ} . Let $J_{\rho}(k)$ denote the declared probability-per-resource objective evaluated with the true fraction. We report

$$\Gamma_{\rho} = \frac{J_{\rho}(\widehat{k})}{J_{\rho}(k_{\rho}^*)}, \quad 0 \leq \Gamma_{\rho} \leq 1, \quad (8)$$

where k_{ρ}^* is the oracle-informed exact- ρ selector used only as a retrospective regret reference. The sweep covers every feasible operational threshold, two attenuation values $\{0, 10^{-3}\}$, two normalized oracle costs $\{1, 4\}$, and all seven multiplicative errors, giving 1,428 conditions. This experiment quantifies estimation sensitivity; it does not claim an estimator-free unknown-solution-count implementation.

3.7. Resource-aware amplitude amplification

For a threshold oracle $O_{\tau}(x) = 1$ iff $C(x) \geq \tau$, let $\rho = \rho_{\tau}$ and define

$$\theta = \arcsin \sqrt{\rho}, \quad (9)$$

and the ideal probability after k standard amplification rounds as

$$P_k = \sin^2((2k+1)\theta). \quad (10)$$

The familiar first continuous peak is

$$k_{\text{peak},0} = \frac{\pi}{4\theta} - \frac{1}{2}. \quad (11)$$

Because Eq. (10) is oscillatory, the first continuous peak is treated only as a descriptive reference. The resource-aware choice is obtained by explicit finite search.

A synthetic attenuation variable $\varepsilon_{\text{model}}$ is introduced only for sensitivity analysis:

$$P_k^{\text{eff}} = \exp[-\varepsilon_{\text{model}}(D_{\text{prep}} + kD_{\text{iter}})] P_k. \quad (12)$$

Here $D_{\text{prep}} = 30$ and $D_{\text{iter}} = 30$ are normalized depth-model units; they are not post-transpilation cycle counts.

To expose oracle cost rather than treating it as free, define a normalized cost

$$R(k) = c_{\text{prep}} + k(c_O + c_D), \quad (13)$$

with $c_{\text{prep}} = 1$, $c_D = 1$, and $c_O \in \{1, 2, 4, 8\}$. The selected number of rounds is

$$k^* = \arg \max_{0 \leq k \leq 50} \frac{P_k^{\text{eff}}}{R(k)}. \quad (14)$$

This objective is dimensionless. It is a probability-per-declared-resource score and must not be read as elapsed-time speedup.

The C^* -normalized calibration experiment evaluates 18 instances, three retrospective quality ratios $q \in \{0.90, 0.95, 1.00\}$, four attenuation levels $\{0, 5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$, and four normalized oracle costs, giving 864 deterministic calibration-sensitivity conditions. The dense 11,934-condition grid retains all three calibration targets. Separately, operational thresholds from Eq. (6) are evaluated under the identical target-hit semantics; exact ρ_τ is used there only as an oracle-informed Tier-I validation reference, not as an operationally available quantity.

3.8. Architecture-informed reversible oracle model and coupled depth

The free normalized oracle-cost axis above is retained as a sensitivity envelope, but it does not represent an implementation. We therefore add a separate architecture-informed logical template for the weighted threshold oracle. Let $m = |E|$, $W = \sum_e w_e$, and

$$b = \lceil \log_2(W + 1) \rceil \quad (15)$$

be the accumulator width. A reusable parity ancilla computes $x_i \oplus x_j$ for one edge, a controlled constant adder accumulates w_{ij} into the b -bit register, and the parity ancilla is uncomputed. After all edges, a reversible comparator writes the event $C(x) \geq \tau$ to a phase flag; the comparator and complete accumulation are then uncomputed.

The template is informed by the linear-depth reversible ripple-carry arithmetic tradition, including one-ancilla addition/comparison constructions and later Clifford+ T gate/depth analyses [25, 26]. It is nevertheless a declared bookkeeping abstraction rather than a claim that those published circuits compile exactly to the counts used here. For reproducibility, the logical ripple-style bookkeeping template assigns $2b$ Toffoli gates, $4b$ CNOT gates, and depth $2b$ to a controlled constant-adder block. Applying the accumulation and its inverse over all

m edges, and computing/uncomputing the comparator, gives

$$N_T^O = 4mb + 4b, \quad (16)$$

$$N_{\text{CNOT}}^O = 8mb + 8m + 8b, \quad (17)$$

$$D_O = 4mb + 8m + 4b + 1. \quad (18)$$

The peak ancilla model is $b + 3$ work qubits, giving $n + b + 3$ logical qubits including the data register. These are architectural logical counts, not transpiled gate counts.

For the diffusion block we use a comparably explicit logical template with $N_T^D = \max(0, 2n - 4)$, $N_{\text{CNOT}}^D = 4n$, and $D_D = \max(1, 2n - 3) + 4$. To obtain one scalar selector while still preserving the separate raw counts, we define the transparent bookkeeping quantity

$$G_{\text{eq}} = N_1 + N_{\text{CNOT}} + 6N_T, \quad (19)$$

where the factor six is only a decomposition-weight convention for a logical Toffoli and is not a backend-native gate claim. To test dependence on that convention, a separate scalarization sensitivity layer recomputes only the mixed bookkeeping projection with $\alpha_T \in \{1, 4, 6, 10\}$ while preserving every raw Toffoli, CNOT, single-qubit, ancilla, and logical-depth count. Fixed overhead in this comparison is anchored to the canonical $\alpha_T = 6$ oracle cost, so changing α_T does not silently rescale the external fixed-work assumption. State preparation contributes $G_{\text{prep}} = n$ single-qubit units and depth one. The coupled model is then

$$R_{\text{arch}}(k) = G_{\text{prep}} + k(G_O + G_D), \quad (20)$$

$$D_{\text{arch}}(k) = 1 + k(D_O + D_D), \quad (21)$$

$$P_{k,\text{arch}}^{\text{eff}} = e^{-\varepsilon_D D_{\text{arch}}(k)} P_k, \quad (22)$$

and k is selected by maximizing $P_{k,\text{arch}}^{\text{eff}}/R_{\text{arch}}(k)$ over $0 \leq k \leq 80$. The synthetic per-logical-depth coefficient is swept over $\varepsilon_D \in \{0, 10^{-5}, 5 \times 10^{-5}, 10^{-4}, 5 \times 10^{-4}\}$. With 51 feasible operational thresholds this produces 255 coupled architecture-informed conditions. This logical layer is not identified with a compiler realization; its purpose is to remove the assumption that oracle cost and iteration depth can vary independently.

Compiler-locked validation layer. To add an executable implementation layer without converting the study into a hardware benchmark, we independently construct weighted-Max-Cut QAOA circuits and a reversible threshold oracle in Qiskit 2.4.2 under Python 3.12.14. Transpilation is locked to basis gates rz/sx/x/cx, optimization level 1, and seed_transpiler=20260901. QAOA circuits for all 18 primary instances and $p = 1, 2, 3$ are routed on a bidirectional line coupling map. The threshold oracle is implemented separately with reusable edge parity, controlled reversible constant addition, an integer comparator, a phase mark, and complete uncomputation; nine representative $\lambda = 0.40$ oracles are basis-decomposed and three sparse representatives are additionally line-routed. A three-vertex weighted semantic test checks all eight computational-basis inputs and requires the work registers to uncompute and the phase to flip iff $C(x) \geq \tau$.

This compiler circuit is an independent executable realization, not an assertion that it is ancilla- or synthesis-identical to the analytical ripple template above. For the same nine $\lambda = 0.40$ representatives, a compiler-native AA cross-check compiles uniform preparation and Grover diffusion in the same locked basis and exhaustively selects $k = 0, \dots, 64$ using P_k divided separately by compiled instruction count and compiled depth; this synthesis-only selector omits a coupling map so that routing remains the separate stress layer above.

A separate fixed-overhead stress test asks whether the zero-round choice is induced by omitting fixed work that is paid once per trial, such as control, readout, verification, or quantum-classical handoff. We therefore define

$$\chi = \frac{C_{\text{fixed}}}{G_O}, \quad R_{\text{arch}}^{\text{fixed}}(k) = C_{\text{fixed}} + G_{\text{prep}} + k(G_O + G_D), \quad (23)$$

and sweep $\chi \in \{0, 0.05, 0.10, 0.25, 0.50, 1, 2\}$ under the same five ε_D values. The exact break-even χ_{BE} is obtained algebraically by comparing each $k > 0$ score with the $k = 0$ score. The primary tier contributes $51 \times 5 \times 7 = 1,785$ rows and the 180 feasible coverage graph/threshold conditions contribute 6,300 rows, for 8,085 fixed-overhead sensitivity conditions. C_{fixed} and χ are synthetic bookkeeping coordinates, not measured back-end costs. This layer replaces neither compilation nor hardware telemetry; its purpose is to remove the assumptions that oracle cost and iteration depth vary independently and that fixed per-trial work is negligible.

3.9. Unknown marked fraction: BBHT schedule

The oracle-informed selectors above use exact Tier-I ρ_τ only as a retrospective reference. To remove that information from the executed stopping policy, we additionally implement the Boyer–Brassard–Høyer–Tapp (BBHT) unknown-solution-count schedule [11]. The control variable starts at $m = 1$ with $\lambda_{\text{BBHT}} = 6/5$. At each trial an integer j is sampled uniformly from the non-negative integers strictly below m , j Grover iterations are applied, the candidate is measured and verified, and after failure m is updated to

$$m \leftarrow \min\{\lambda_{\text{BBHT}} m, \sqrt{N}\}. \quad (24)$$

The schedule therefore depends on $N = 2^n$ but not on the number of marked states. For the logical resource comparison, each randomized- j trial is charged the same preparation, iteration, attenuation, and fixed-overhead terms used in $R_{\text{arch}}^{\text{fixed}}$. Before saturation one attempt is made at each growth stage; after $m = \sqrt{N}$ is reached, the saturated randomized trial repeats until success. We evaluate the resulting expected trials, Grover iterations, and gate-equivalent resource-to-success exactly by summing the pre-saturation survival probabilities and the final geometric tail. Exact ρ_τ enters only this retrospective expected-cost evaluation, never the BBHT growth schedule itself. The comparison is reported against repeated uniform sampling and the oracle-informed exact- ρ architecture selector for $\varepsilon_D \in \{0, 10^{-4}\}$ and $\chi \in \{0, 0.25\}$.

3.10. QAOA protocol

All 18 primary exact instances enter the QAOA study; no graph seed is discarded for the inferential layer. QAOA uses the standard weighted Max-Cut cost phase and an X mixer. For depth p , the state is

$$|\psi(\gamma, \beta)\rangle = \prod_{\ell=1}^p U_B(\beta_\ell) U_C(\gamma_\ell) |+\rangle^{\otimes n}. \quad (25)$$

Depths $p \in \{1, 2, 3\}$ are optimized independently. For each instance and depth, five optimizer seeds are used in the primary analysis, yielding $18 \times 3 \times 5 = 270$ runs. A separate initialization-stability audit expands the same deterministic instance-conditioned stream to seeds 0–19, yielding $18 \times 3 \times 20 = 1,080$ L-BFGS-B runs; seeds 0–4 are therefore an exact nested subset of the expanded audit. Initial γ values are drawn uniformly from $[0, 2\pi]$ and β values from $[0, \pi]$. A versioned deterministic initialization stream is conditioned on graph identity, depth, and optimizer seed, so the same external seed does not reuse an identical parameter vector across different graph instances. L-BFGS-B is allowed at most 60 iterations. The objective is evaluated from the exact statevector expectation, so optimization-shot noise is absent. After optimization, both the exact optimum-state probability and the common threshold probabilities $P_{\tau_q} = \Pr[C(x) \geq \tau_q]$ are computed directly from the statevector for $q = 0.90, 0.95, 1.00$. A 4096-shot optimum-state sample is retained as a measurement-style diagnostic but is not used to optimize the circuit.

The main QAOA performance quantity is the approximation ratio

$$\alpha = \frac{\mathbb{E}[C(x)]}{C^*}. \quad (26)$$

For each instance and depth, we report the median and interquartile range across the five optimizer initializations. Aggregated results then summarize the distribution of these instance-level medians.

For optimizer-family robustness, a separate paired ablation uses L-BFGS-B and COBYLA from exactly the same initial parameter vector at every (instance, p , seed) tuple. Each optimizer receives a strict cap of 128 calls to the exact statevector expectation objective. Calls used internally by numerical finite-difference gradients count against the L-BFGS-B cap; direct COBYLA objective calls count in the same way. Early optimizer convergence is permitted, so the cap is a maximum rather than a requirement to consume all 128 evaluations. The final statevector metrics are computed once after optimization and are not charged to this optimization cap. The design therefore contains $18 \times 3 \times 5 \times 2 = 540$ paired optimizer runs. Host optimization time is logged only as a diagnostic and is not treated as a portable or physical quantum runtime metric.

A paired objective-alignment sensitivity uses the same 18 instances, depths, five instance-conditioned starts, L-BFGS-B, and strict 128-evaluation cap, replacing only expected-cut training by direct optimization of $\Pr[C(x) \geq \tau_{0.40}^{\text{op}}]$; inference collapses starts within instance before the same Wilcoxon/Holm and bootstrap checks.

3.11. Instance-clustered statistical analysis

The inferential layer treats benchmark instances, rather than optimizer starts, as the paired experimental units. For the primary QAOA depth study, the five L-BFGS-B starts are first collapsed to an instance-level median approximation ratio at each p . The resulting 18 matched instance values per depth are tested with a Friedman test across $p = 1, 2, 3$. Pairwise depth differences are then assessed with two-sided Wilcoxon signed-rank tests and Holm family-wise correction across the three depth contrasts. For the finite-budget optimizer ablation, the five paired COBYLA-minus-L-BFGS-B differences are first collapsed to one median difference for each instance and depth; a two-sided Wilcoxon signed-rank test is then applied at each depth, with Holm correction across the three optimizer tests. We report matched-pairs rank-biserial correlation and a percentile 95% bootstrap interval for each median paired difference using 20,000 resamples of the 18 instances. The suite is deterministic and small, so these tests describe consistency within the fixed benchmark; they are not interpreted as population-level inference over a graph-generating distribution.

3.12. Classical references

The classical layer contains one canonical approximation reference and three transparent heuristics. The canonical reference is the Goemans–Williamson semidefinite relaxation [27]. For a weighted graph with Laplacian L , the primal relaxation is

$$\max_{X \geq 0, \text{diag}(X)=I} \frac{1}{4} \text{tr}(LX), \quad (27)$$

and the corresponding dual is $\min_y \sum_i y_i$ subject to $\text{Diag}(y) - L/4 \geq 0$. The implementation uses an n -dimensional Gram factor, so the factorization can represent every feasible $n \times n$ primal matrix. Deterministic multistart block-coordinate ascent updates each unit vector against its weighted neighbour field. The numerical primal solution is not accepted on optimization quality alone: every restart also produces a dual candidate, which is shifted uniformly by the minimum amount required to make the dual slack numerically positive semidefinite. The largest primal value and smallest dual-feasible bound across 12 starts must close to within 10^{-6} ; otherwise the workflow fails. A certified factor is then rounded with 4096 deterministic-seed random hyperplanes. Exact C^* is not used by the SDP solver or rounding process and appears only in retrospective exact-tier scoring.

The heuristic references are retained for a complementary finite-search view. First, the original sanity baseline uses 256 random starting bitstrings per graph and greedily applies the best strictly improving one-bit flip until a local optimum is reached. Because its number of objective evaluations is path dependent, this hill climber is not used as a budget-matched reference. Second, simulated annealing is run 64 times per instance with a geometric temperature schedule whose initial scale is derived from the maximum weighted degree. Third, a one-flip tabu search is run 64 times per instance with a short deterministic tabu tenure and aspiration for moves that improve the run-best value. Rather than relying on one arbitrarily large cap, simulated annealing and tabu search are

evaluated at identical per-run objective-evaluation budgets $B \in \{64, 128, 256, 512, 1024, 4096\}$. Each budget is a complete budget-specific run; in particular, the annealing temperature schedule is rescaled to the declared budget rather than obtained by truncating a 4096-evaluation trajectory. The 4096 setting is retained only as the high-budget endpoint and for continuity with the earlier fixed-budget comparison. Their search logic does not query C^* or optimum-state identities; exact enumeration is used only afterward to compute approximation ratios, optimum-hit indicators, and threshold-hit fractions. Objective evaluations need not correspond to unique states, and the budget is a within-classical fairness device that is not equated to QAOA circuit evaluations or amplitude-amplification resources.

3.13. Reproducibility protocol

All primary and coverage graphs, exact optima, threshold definitions, analytical sensitivity rows, coverage stress-test rows, common target-metric rows, QAOA runs, QAOA summaries, the certified SDP/rounding reference, and raw and summarized classical-reference results are generated from one source implementation. The run manifest records the suite definition, optimizer settings, evidence classes, Python, NumPy, and SciPy versions, and SHA-256 hashes of the principal machine-readable artifacts. The submission reproducibility layer additionally pins Python 3.13.5 in `.python-version`, exact direct dependencies in `requirements.txt`, and the validated dependency closure in `requirements.lock`. A fail-closed `src/preflight.py` validator checks manifest hashes, declared CSV row counts, canonical file locations, package-version drift, LaTeX environment balance, and the manuscript claim boundary before the submission package is synchronized. Tables and figures are rendered only from machine-generated outputs. No manuscript target value is hard-coded into the computational path.

OpenAI ChatGPT was used as an AI-assisted tool during project development to support code drafting, test design, manuscript organization, language editing, literature-navigation support, and consistency checking. The final reviewer-driven audit and revision on 1 September 2026 used ChatGPT with the GPT-5.6 Sol model. Earlier ChatGPT-assisted development sessions were not consistently logged at the exact model-version level, so no retrospective model identifier is inferred for those sessions. The author executed and reviewed the computational workflow, inspected the resulting artifacts, independently verified the reported numerical evidence through the declared tests and preflight checks, checked cited sources against publisher or bibliographic records, and retained responsibility for all methodological choices, interpretations, references, and final text. AI-generated suggestions were not treated as empirical or computational evidence unless they were reproduced by the project workflow.

3.14. Three-tier scaling and robustness design

To separate exact evidence from extrapolative modeling, the benchmark uses three tiers. Tier I contains the 18-instance

primary exact suite with $n \leq 12$ plus the separately labeled 63-instance exact coverage stress test; all 18 primary instances enter the 18-instance QAOA inference, while the additional coverage instances do not. Tier II is a host-side simulation-scaling diagnostic at $n \in \{8, 10, 12, 14, 16, 18, 20\}$ using exact enumeration and one fixed-parameter $p = 1$ statevector evaluation; its timings characterize only the reference Python/NumPy implementation and are not quantum-runtime measurements. Tier III is analytical only: abstract search spaces with $n_{\text{model}} \in \{12, 16, 20, 24\}$ and target-event multiplicities $M_\tau \in \{1, 4, 16\}$ are swept over the same resource-uncertainty coordinates. These Tier-III rows are not asserted to be Max-Cut ground truth and do not encode optimum bitstrings; they isolate the behavior of the selector as a threshold-satisfying event becomes rarer.

The dense robustness grid uses all $q \in \{0.90, 0.95, 1.00\}$ thresholds, 13 attenuation values from 0 to 0.003, and 17 logarithmically spaced normalized oracle costs from 0.5 to 16. On each exact instance, the same finite-set selector is evaluated at every grid point, giving $18 \times 3 \times 13 \times 17 = 11,934$ conditions. We additionally report the baseline-relative statistic

$$\eta_{\text{res}} = \frac{R(k^*)/P_{k^*}^{\text{eff}}}{R(0)/P_0^{\text{eff}}}, \quad (28)$$

which is the expected declared resource per successful hit relative to no amplification under the same model. Values below one indicate improvement only under the stated normalized objective; they are not speedup factors.

4. Results

4.1. Exact suite characteristics

The 18-instance suite has exact optimum cut values from 31 to 175. For the $q = 0.95$ reference threshold, the marked fraction ρ_τ ranges from 1.9531×10^{-3} to 6.25×10^{-2} . Varying the target ratio changes the marked fraction and therefore the amplification period without changing the graph itself; this provides a direct target-quality sensitivity axis.

4.2. Operational-threshold validation

The operational construction exposes a distinction hidden by C^* -normalized targets. At $\lambda = 0.25$ and 0.40, all 18 declared thresholds are feasible under retrospective exact validation. At $\lambda = 0.55$, 15 of 18 are feasible and three lie above the exact optimum; those three are retained as infeasible declarations rather than silently repaired. Among feasible cases, the retrospective median $\tau_\lambda^{\text{op}}/C^*$ is 0.790, 0.865, and 0.933 for $\lambda = 0.25, 0.40, 0.55$, respectively. The corresponding median exact marked fractions are 0.1816, 0.0566, and 0.0156. Thus the operational level controls strictness without requiring C^* , but a graph-derived upper bound does not guarantee that every interpolated target is attainable.

The dense 0.05-spaced audit shows the same trend outside the three declared points. Primary feasibility is 18/18 at $\lambda = 0$, remains 18/18 at 0.25 and 0.40, falls to 15/18 at 0.55, 5/18 at

Table 3: Median target-hit probability for operational thresholds on the 18 primary instances, restricted to thresholds found feasible by exact Tier-I validation. The AA reference uses exact ρ_τ only for validation-stage stopping-depth selection; method-specific resource units remain unmatched.

λ	feas.	uniform	AA ref.	HC	QAOA $p = 1$	$p = 2$	$p = 3$
0.25	18/18	0.1816	0.9383	1.0000	0.1930	0.2784	0.3081
0.40	18/18	0.0566	0.8488	0.9707	0.0683	0.1187	0.1143
0.55	15/18	0.0156	0.8229	0.8633	0.0156	0.0252	0.0547

0.75, and 2/18 at 1.00. In connected validation the corresponding counts are 45/45, 45/45, 45/45, 36/45, 12/45, and 0/45. Across every feasible dense-sweep condition, the zero-fixed-overhead, zero-attenuation architecture-informed selector still returns $k^* = 0$. The full 21-level machine-readable curve is retained in the reproducibility package.

Across the 18 primary instances shared by all methods, the operational target event is feasible on 18/18, 18/18, and 15/18 instances for the three levels. Table 3 reports target-hit probabilities only over the feasible instances. The amplitude-amplification column is intentionally labeled as an *AA reference*: its stopping depth still uses the exact Tier-I ρ_τ after threshold construction. Removing that marked-fraction knowledge is treated as a separate robustness problem rather than being hidden inside the threshold definition.

4.3. Expanded seed and topology coverage

The separate exact coverage tier contains 45 random weighted instances and 18 structured weighted instances, 63 in total. The operational threshold remains feasible for every coverage instance at $\lambda = 0.25$, on 62/63 instances at $\lambda = 0.40$, and on 55/63 instances at the stricter $\lambda = 0.55$ level. Family-level behavior is not identical: the $\lambda = 0.55$ target is feasible on 39/45 random weighted graphs, 6/6 cycles, 6/6 3-regular circulants, and 4/6 two-community graphs. Retrospective threshold quality also shifts by topology; at $\lambda = 0.55$, the median τ/C^* is 0.935 for the random weighted family but 0.785 for cycles, 0.809 for 3-regular circulants, and 0.944 for the feasible two-community cases. These differences show why the operational level should not be interpreted as a topology-invariant approximation ratio.

The architecture-informed conclusion is more stable within this micro-instance coverage. Across the 180 feasible operational graph/threshold combinations and two declared attenuation values $\varepsilon_D \in \{0, 10^{-3}\}$, all 360 coverage conditions select $k^* = 0$ under the same gate-equivalent probability-per-resource objective. This extends the negative configuration result beyond the original 18 random instances but remains an exact micro-instance statement under one reversible logical template, not a large-scale or hardware claim. The 64-start hill-climbing coverage diagnostic is also strong: family-level median ratios are 1.000 for random weighted graphs, 0.958 for cycles, 1.000 for 3-regular circulants, and 0.977 for two-community graphs.

4.4. Structural audit and connected validation

The graph-structure audit confirms that the original fixed benchmark contains identifiable structural degeneracies rather

Table 4: Expanded exact seed/topology coverage. Feasibility and τ/C^* are retrospective exact-validation quantities; HC denotes the 64-start hill-climbing coverage diagnostic. The QAOA inferential suite remains restricted to the 18 primary rows and is not enlarged by the additional coverage rows.

family	λ	feas.	med. τ/C^*	med. ρ_τ	med. HC AR
random weighted	0.25	45/45	0.774	0.1875	1.000
random weighted	0.40	44/45	0.859	0.0625	1.000
random weighted	0.55	39/45	0.935	0.0156	1.000
cycle	0.25	6/6	0.633	0.2705	0.958
cycle	0.40	6/6	0.710	0.1416	0.958
cycle	0.55	6/6	0.785	0.0762	0.958
3-regular circulant	0.25	6/6	0.654	0.1953	1.000
3-regular circulant	0.40	6/6	0.732	0.0801	1.000
3-regular circulant	0.55	6/6	0.809	0.0322	1.000
two-community	0.25	6/6	0.805	0.1973	0.977
two-community	0.40	6/6	0.876	0.0413	0.977
two-community	0.55	4/6	0.944	0.0178	0.977

Table 5: Structural audit of the exact graph tiers. The connected validation tier is conditioned on connected, non-bipartite topology; it does not replace the original primary benchmark.

tier	inst.	disconnected	bipartite	isolated-bearing	med. M_{opt}
primary	18	5	2	5	2.0
expanded coverage	63	10	13	8	2.0
connected validation	45	0	0	0	2.0

than silently assuming all random instances are connected. Five of the 18 primary graphs are disconnected, two are bipartite, and five contain at least one isolated vertex. Across the 63-instance expanded coverage tier, 10 graphs are disconnected and 13 are bipartite. Table 5 records these counts explicitly. The separate 45-instance validation tier contains no disconnected, bipartite, or isolated-vertex cases by construction.

The main resource conclusion survives this structural control. In the connected validation tier the operational targets are feasible on 45/45, 45/45, and 36/45 instances at $\lambda = 0.25, 0.40, 0.55$, with median retrospective τ/C^* values 0.794, 0.886, and 0.946. Across the 126 feasible graph/threshold combinations and both $\varepsilon_D \in \{0, 10^{-3}\}$, all 252 zero-fixed-overhead architecture-coupled conditions still select $k^* = 0$. The fixed-overhead boundary remains nontrivial: at zero attenuation the median break-even ratio over all feasible connected-validation conditions is $\chi_{BE} = 0.121$ (IQR 0.065–0.190); 90.5% select nonzero amplification at $\chi = 0.25$, and all do so at $\chi = 0.50$. Thus the zero-round result is not explained solely by disconnected or bipartite graphs, while the same fixed-overhead qualification remains necessary.

4.5. Marked-fraction misspecification

The marked-fraction audit shows that the discrete resource-aware selector is locally stable to moderate marked-fraction error on this exact suite. With $\widehat{\rho}/\rho = 0.90$ or 1.10, the median retained true objective is 1.000; the worst observed retention is 0.954 and 0.960, respectively. At -25% and $+25\%$ error, the median is still 1.000, while worst-case retention is 0.859 and 0.853. A factor-of-two underestimate is materially more damaging: median retention is 0.857, median $|\Delta k| = 1$, and the worst case retains only 0.505 of the oracle-informed objective.

Table 6: Connected, non-bipartite exact validation through $n = 14$. The zero-overhead column reports the fraction selecting $k > 0$ at $\varepsilon_D = 0$; χ_{BE} is the exact fixed-overhead break-even ratio in oracle-cost units.

λ	feasible	med. τ/C^*	med. ρ_τ	nonzero k @ $\chi = 0$	med. χ_{BE}
0.25	45/45	0.794	0.1400	0.000	0.208
0.40	45/45	0.886	0.0337	0.000	0.115
0.55	36/45	0.946	0.0067	0.000	0.053

Table 7: Architecture-informed reversible weighted-Max-Cut threshold-oracle resources. Counts follow the declared ripple-style logical template and are not post-transpilation metrics.

n	inst.	m range	b range	log. qubits	med. Toffoli _O	med. D_O
8	6	7–21	6–7	17–18	420	533
10	6	11–34	6–8	19–21	644	821
12	6	16–50	7–8	22–23	1088	1353

A 50% overestimate has median retention 0.985 and worst-case retention 0.648.

The asymmetry is consistent with the stopping rule: underestimating the marked fraction tends to select a deeper amplification schedule, which can overshoot the useful region once the true threshold event is less rare than assumed. These results remove the hidden implication that exact ρ_τ is harmless, while keeping the exact- ρ selector as an explicitly labeled upper-reference rather than a deployable assumption.

4.6. Architecture-informed oracle resources and coupled selection

The reversible threshold-oracle template is substantially more expensive than the abstract free-oracle coordinates used in the sensitivity envelope. Across the six instances at each size, the median modeled oracle Toffoli count and logical oracle depth are 420 and 533 for $n = 8$, 644 and 821 for $n = 10$, and 1,088 and 1,353 for $n = 12$. Depending on graph density and accumulator width, the modeled total logical-qubit requirement ranges from 17–18, 19–21, and 22–23 qubits at those three sizes, respectively. Table 7 reports the full size-level summary.

When these logical gate-equivalent costs are coupled to the corresponding oracle/diffusion depths with $C_{\text{fixed}} = 0$, the result changes qualitatively. Across all 51 feasible operational thresholds and all five ε_D values, every one of the 255 coupled conditions selects $k^* = 0$. This remains true even at $\varepsilon_D = 0$, so under the zero-fixed-overhead convention the rejection of amplification is driven first by the declared logical oracle cost rather than by attenuation. Consequently, the earlier normalized-cost sweeps must be read as uncertainty maps that identify how the selector behaves if the oracle is assigned a chosen relative cost; they are not evidence that the weighted-threshold oracle remains resource-favorable after an explicit reversible construction is charged. Table 8 records the endpoint summaries.

The independent compiler layer supports the same qualitative resource warning without equating the two models. Across the 54 line-routed QAOA circuits, median compiled depth is 78.0, 180.5, and 260.5 for $p = 1, 2, 3$, with median CX counts 97.0, 234.5, and 348.0. The executable weighted-threshold

Table 8: Coupled architecture-informed selector at the attenuation endpoints. η_{arch} is resource per target hit relative to the no-amplification baseline under the same logical bookkeeping model.

λ	feasible	ε_D	median k^*	zero-round frac.	median η_{arch}
0.25	18	0e+00	0.0	1.000	1.000
0.25	18	5e-04	0.0	1.000	1.000
0.40	18	0e+00	0.0	1.000	1.000
0.40	18	5e-04	0.0	1.000	1.000
0.55	15	0e+00	0.0	1.000	1.000
0.55	15	5e-04	0.0	1.000	1.000

Table 9: Fixed per-trial overhead sensitivity. $\chi = C_{fixed}/G_O$ and χ_{BE} is the exact overhead ratio at which some $k > 0$ first exceeds the no-amplification probability-per-resource score. These are logical-model sensitivity coordinates, not measured backend costs.

scope	ε_D	med. χ_{BE}	IQR χ_{BE}	nonzero@0.25	nonzero@0.50
primary	0e+00	0.163	0.098–0.225	0.784	1.000
primary	5e-04	0.306	0.230–0.414	0.353	0.863
coverage	0e+00	0.168	0.103–0.239	0.772	1.000
coverage	5e-04	0.301	0.224–0.427	0.389	0.844

oracle passes the 8/8-basis-state semantic check. Across nine representative basis-decomposed operational oracles, median compiled depth is 110,317 and median CX count is 53,794. A matched abstraction-gap ledger compares only like-named depth/qubit coordinates rather than equating gate alphabets: the executable-oracle depth is a median 140.5 $\times$ the analytical ripple-template depth (IQR 138.0–168.0 $\times$), while compiled qubit count is a median 1.61 $\times$ the analytical logical-qubit estimate. For sparse $n = 8, 10, 12$ representatives routed on the same line topology, depths are 41,539/65,177/130,085 and CX counts are 30,655/48,636/97,268. These are compiler-locked synthetic-topology counts, not device duration, fidelity, or physical-resource measurements; the large abstraction gap is itself evidence that the analytical template should be interpreted as configuration bookkeeping rather than a compiler-count predictor. On those nine $\lambda = 0.40$ representatives, the compiler-native AA cross-check still selects $k^* = 0$ in 9/9 cases under both compiled instruction count and compiled depth; median fixed-per-trial break-even coordinates are 0.148 and 0.148 times one compiler-native iteration cost, respectively. These coordinates have a different denominator from analytical C_{fixed}/G_O and are not treated as metric-equivalent.

The fixed-overhead stress test changes the interpretation without changing the zero-overhead result. At $\varepsilon_D = 0$, the median exact break-even ratio is $\chi_{BE} = 0.163$ (IQR 0.098–0.225) over the 51 primary feasible thresholds and 0.168 (0.103–0.239) over the 180 feasible coverage thresholds. At $\chi = 0.25$, nonzero amplification is selected in 78.4% and 77.2% of primary and coverage conditions, respectively; at $\chi = 0.50$, every zero-attenuation condition selects $k^* > 0$. At the strongest tested attenuation, $\varepsilon_D = 5 \times 10^{-4}$, the median break-even ratios rise to 0.306 and 0.301; the nonzero fractions at $\chi = 0.50$ are still 86.3% and 84.4%. Thus “all $k^* = 0$ ” is a valid result only for the explicitly declared negligible-fixed-overhead boundary, not a general statement about hybrid workflows. Table 9 and Fig. 1 expose this configuration boundary.

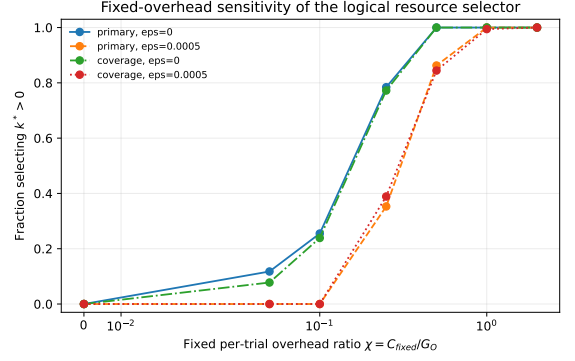


Figure 1: Fraction of feasible conditions selecting nonzero amplitude amplification as fixed per-trial overhead increases. The boundary is shown for the primary and coverage tiers at zero and maximum tested synthetic attenuation.

4.7. Scalarization sensitivity of the logical resource score

The qualitative zero-fixed-overhead conclusion does not depend on the factor-six Toffoli convention. At $\varepsilon_D = 0$ and $\chi_6 = 0$, every feasible primary and connected-validation condition selects $k^* = 0$ for all $\alpha_T \in \{1, 4, 6, 10\}$. The phase boundary, however, shifts as expected when the same external fixed work is compared against a differently weighted oracle. In the primary tier, the median break-even ratio expressed in units of the canonical $\alpha_T = 6$ oracle cost rises from 0.064 at $\alpha_T = 1$ to 0.124, 0.163, and 0.243 at $\alpha_T = 4, 6, 10$. The connected-validation medians are 0.047, 0.091, 0.121, and 0.180.

At a fixed canonical overhead $\chi_6 = 0.25$ and zero attenuation, the fraction selecting nonzero amplification in the primary tier decreases from 1.000 at $\alpha_T = 1$ to 0.961, 0.784, and 0.549 as the Toffoli weight increases; the connected-validation fractions are 1.000, 0.976, 0.905, and 0.667. Thus the existence of a zero-overhead no-amplification boundary is robust to this scalarization, while the location of the fixed-overhead transition is not. The full zero-attenuation scalarization summary is retained in the reproducibility package.

4.8. Unknown marked fraction via BBHT

The BBHT layer removes exact ρ_T from the executed growth schedule and exposes the cost of doing so. At zero attenuation and zero fixed overhead, BBHT is never preferred to repeated uniform sampling in either exact tier: its median resource-to-success ratio relative to uniform is 54.651 in the primary tier and 57.797 in the connected-validation tier. This is consistent with the architecture-informed result above: when preparation/fixed work is negligible, randomized Grover iterations add a large reversible-oracle burden to a cheap $k = 0$ trial.

The comparison changes once fixed per-trial work is admitted. At $\chi = 0.25$ and $\varepsilon_D = 0$, the median BBHT/uniform resource ratio falls to 0.769 in the primary tier and 0.643 in connected validation; BBHT is resource-preferred to uniform repetition in 76.5% and 90.5% of feasible threshold conditions, respectively. The information-free schedule still pays a premium relative to the oracle-informed exact- ρ selector: the corresponding median BBHT/oracle-informed ratios are 1.170 and

Table 10: BBHT unknown-solution-count comparison. BBHT/uniform and BBHT/oracle are expected logical resource-to-success ratios. The BBHT schedule does not receive ρ_τ ; exact ρ_τ is used only for retrospective expected-cost evaluation.

tier	ε_D	χ	med. BBHT/uniform	BBHT better frac.	med. BBHT/oracle
primary	0e+00	0.00	54.651	0.000	54.651
primary	0e+00	0.25	0.769	0.765	1.170
primary	1e-04	0.25	1.043	0.471	1.230
connected val.	0e+00	0.00	57.797	0.000	57.797
connected val.	0e+00	0.25	0.643	0.905	1.247
connected val.	1e-04	0.25	1.146	0.349	1.502

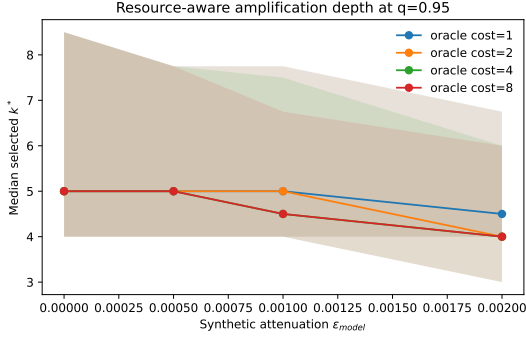


Figure 2: Median selected amplitude-amplification depth across 18 exact Max-Cut instances. Shaded regions show the interquartile range. Both attenuation and oracle cost are normalized sensitivity variables, not device measurements.

1.247. With $\varepsilon_D = 10^{-4}$ at the same fixed overhead, BBHT is better than uniform in only 47.1% of primary and 34.9% of connected-validation conditions. Thus unknown marked fraction does not erase the fixed-overhead phase boundary, but it narrows the region in which amplification is resource-preferred. Table 10 summarizes this layer.

4.9. Amplitude-amplification sensitivity

Figure 2 and Table 11 summarize the declared $q = 0.95$ threshold reference over the 18 exact instances. At unit normalized oracle cost, the median selected k^* is 5.0 at $\varepsilon_{\text{model}} = 0$, remains 5.0 at 5×10^{-4} and 10^{-3} , and decreases to 4.5 at 2×10^{-3} . With normalized oracle cost 8, the corresponding medians are 5.0, 5.0, 4.5, and 4.0. These depths are materially shallower than exact-optimum marking because the threshold oracle marks a larger, explicitly quality-conditioned set.

The oracle-cost effect remains instance dependent. The important change is semantic rather than merely numerical: k^* is now selected for a declared quality threshold, so a reviewer can reproduce the marked-set rule from the cut-value comparator without assuming that the identities of optimum states are known in advance. The normalized oracle-cost sweep still does not substitute for compiling that comparator into reversible arithmetic.

4.10. Dense resource-uncertainty regions

The threshold-aware dense uncertainty sweep contains 11,934 exact-instance conditions. At $q = 0.90$, the median selected depth decreases from 3.0 to 2.0 across the attenuation endpoints, with the lower quartile reaching zero at

Table 11: Aggregate amplitude-amplification sensitivity at the declared $q = 0.95$ quality threshold. The complete machine-readable output also contains $q = 0.90$ and $q = 1.00$.

$\varepsilon_{\text{model}}$	oracle cost	median k^*	IQR k^*	median η_{res}
0.0000	1	5.0	4.0–8.5	0.141
0.0000	2	5.0	4.0–8.5	0.206
0.0000	4	5.0	4.0–8.5	0.334
0.0000	8	5.0	4.0–8.5	0.591
0.0005	1	5.0	4.0–7.8	0.152
0.0005	2	5.0	4.0–7.8	0.222
0.0005	4	5.0	4.0–7.8	0.360
0.0005	8	5.0	4.0–7.8	0.637
0.0010	1	5.0	4.0–7.8	0.164
0.0010	2	5.0	4.0–7.8	0.239
0.0010	4	4.5	4.0–7.5	0.387
0.0010	8	4.5	4.0–6.8	0.684
0.0020	1	4.5	4.0–6.8	0.189
0.0020	2	4.0	4.0–6.8	0.273
0.0020	4	4.0	3.0–6.0	0.440
0.0020	8	4.0	3.0–6.0	0.776

$\varepsilon_{\text{model}} = 0.003$; the median η_{res} changes from 0.407 to 0.529. At $q = 0.95$, the median selected depth decreases from 5.0 to 4.0 and the median η_{res} changes from 0.265 to 0.395. At $q = 1.00$, the median depth decreases from 9.0 to 6.0 while median η_{res} changes from 0.149 to 0.312. Across all targets and resource coordinates, 13.3% of conditions select $k^* = 0$. Thus target strictness is itself a first-order configuration variable rather than a fixed background assumption.

4.11. Host simulation boundary and analytical large-space tier

The reference implementation exhibits the expected rapid growth in classical simulation cost. At the two endpoint sizes, exact construction takes approximately 1 ms at $n = 8$ and 0.70 s at $n = 20$ on the recorded host, while a single fixed-parameter $p = 1$ statevector evaluation rises from approximately 1 ms to 0.49 s. The two principal NumPy vectors used in this diagnostic grow from 0.004 MiB to 16 MiB. These timings define the evidence boundary of this implementation only; they do not measure quantum performance. The corresponding scaling figure is retained in the reproducibility package.

The analytical Tier-III study demonstrates why resource assumptions become increasingly consequential when threshold-satisfying target events are rare. With one target-event state and zero attenuation, the median selected depth over the oracle-cost grid grows from 37 rounds for a 2^{12} search space to about 2,386 rounds for 2^{24} . At $\varepsilon_{\text{model}} = 0.003$, however, the median collapses to about 10 rounds across the larger modeled spaces. This is not a scalability claim for Max-Cut or a device forecast; it is a controlled sensitivity result showing that an attenuation penalty can dominate the ideal Grover-period scaling in the declared objective. The supporting large-tier figure is retained in the reproducibility package.

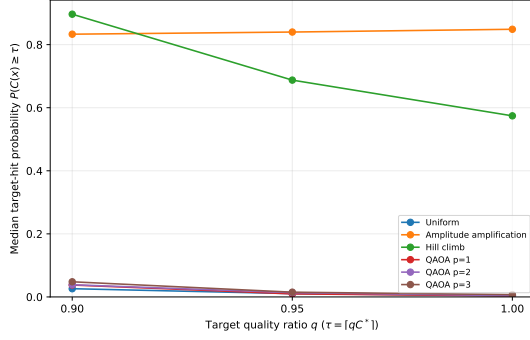


Figure 3: Core target-hit probability view on the 18 primary instances. The target event is identical, $C(x) \geq \tau_q$, while method-specific execution costs remain deliberately unmerged. The fixed-budget simulated-annealing and tabu-search references are reported separately in the classical-reference table and machine-readable ledger.

Table 12: Median target-hit probability for the core quantum/uniform/hill-climb view on the 18 primary instances. AA denotes the query-level reference operating point with zero synthetic attenuation and unit normalized oracle cost; stronger fixed-budget classical references are reported separately.

target	uniform	AA	hill climb	QAOA $p = 1$	$p = 2$	$p = 3$
0.90	0.0261	0.8331	0.8965	0.0378	0.0387	0.0480
0.95	0.0105	0.8399	0.6875	0.0097	0.0136	0.0151
1.00	0.0039	0.8487	0.5742	0.0033	0.0041	0.0074

4.12. Benchmark-normalized calibration targets across methods

As a complementary exact-tier calibration, the benchmark reports the same event

$$P_{\tau_q} = \Pr[C(x) \geq \tau_q] \quad (29)$$

for uniform sampling, the reference amplitude-amplification operating point ($\varepsilon_{\text{model}} = 0$, unit normalized oracle cost), multistart hill climbing, fixed-budget simulated annealing, fixed-budget tabu search, and each tested QAOA depth. This is a common success target, not a claim that a QAOA evaluation, a classical objective evaluation, and an amplitude-amplification resource unit have equal physical cost.

Across all 18 primary instances, the median target-hit probabilities at $q = 0.95$ are 0.0105 for uniform sampling, 0.8399 for the query-level amplitude-amplification reference, 0.6875 for hill climbing, 1.000 for simulated annealing, 1.000 for tabu search, and 0.0115, 0.0123, and 0.0124 for QAOA depths $p = 1, 2, 3$. The two fixed-budget classical heuristics also have median target-hit fraction 1.000 at $q = 0.90$ and $q = 1.00$. Thus the stronger classical references dominate the common target-hit statistic on these micro-instances. The probabilities still do not establish comparative runtime advantage because the method-specific resource units are intentionally not merged.

4.13. QAOA depth and optimizer variability

Of the 270 QAOA optimization runs, 269 return a successful optimizer convergence flag. The aggregate results are shown in Table 13 and Figure 4. The median of the 18 instance-level

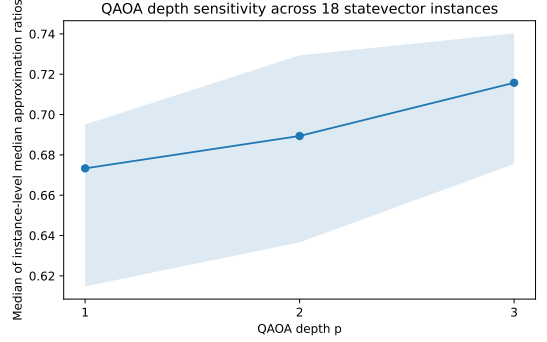


Figure 4: QAOA depth sensitivity across 18 exact instances. The line shows the median of instance-level median approximation ratios; the band shows the interquartile range across instances. Each instance-level median is computed from five independent optimizer initializations.

Table 13: Aggregate ideal-statevector QAOA results. Each depth contains 18 instances and five optimizer starts per instance.

p	instances	runs	median instance AR	median instance P_{opt}
1	18	90	0.673	0.0033
2	18	90	0.689	0.0041
3	18	90	0.716	0.0074

median approximation ratios is 0.673 at $p = 1$, 0.689 at $p = 2$, and 0.716 at $p = 3$. Under the instance-conditioned initialization protocol the fixed-suite central statistic increases at each tested depth.

The median of instance-level median optimum-state probabilities is 0.00327, 0.00407, and 0.00737 for $p = 1, 2, 3$. These probabilities have wide between-instance and between-seed variation. 15 of the 18 instances attain their best median approximation ratio at $p = 3$, whereas 3 favor $p = 2$. This fixed-suite pattern does not establish a universal depth rule beyond the tested benchmark.

The best single run in the complete statevector sweep occurs for instance n8_d25_s42 at $p = 3$ and optimizer seed 0, with approximation ratio 0.9455 and exact optimum-state probability 0.6193. This best-run value is retained in the raw ledger but is not used as the representative result.

The five-start primary result is not used as the only initialization sample. In the nested 20-start audit, the median of instance-level median approximation ratios is 0.661, 0.691, and 0.707 at $p = 1, 2, 3$. The corresponding Friedman statistic is 26.778 ($p = 1.53 \times 10^{-6}$). All three paired depth contrasts remain positive after collapsing 20 starts within each instance: $p = 2$ versus $p = 1$ has median $\Delta = +0.0126$ (95% bootstrap interval [0.0046, 0.0305], Holm $p = 0.00033$), $p = 3$ versus $p = 1$ has +0.0334 ([0.0217, 0.0653], Holm $p = 0.00002$), and $p = 3$ versus $p = 2$ has +0.0169 ([0.0062, 0.0337], Holm $p = 0.00011$). Thus the depth ordering is not an artifact of using only five initializations on this fixed suite.

The equal-cap optimizer ablation gives a different but complementary view. Under the 128-objective-evaluation cap, the median of the 18 instance-level median approximation ratios for L-BFGS-B is 0.673, 0.702, and 0.696 at $p = 1, 2, 3$,

Table 14: Twenty-start QAOA initialization-stability audit. Seeds 0–4 are the exact five-start primary subset; inference again collapses starts within each of the 18 fixed benchmark instances before testing.

p	starts/inst.	runs	median inst. AR	median inst. P_{opt}
1	20	360	0.661	0.0039
2	20	360	0.691	0.0039
3	20	360	0.707	0.0095

20-start depth contrasts		95% CI	P_{Holm}	n
contrast	median Δ			
p2-p1	+0.0126	[+0.0046,+0.0305]	0.00033	18
p3-p1	+0.0334	[+0.0217,+0.0653]	0.00002	18
p3-p2	+0.0169	[+0.0062,+0.0337]	0.00011	18

Table 15: Paired ideal-statevector QAOA optimizer robustness under a common cap of 128 exact expectation-objective evaluations per run. “Budget-exhaust.” is the fraction of runs that end at the cap without an earlier successful optimizer termination.

optimizer	p	runs	median inst. AR	median evals	budget-exhaust.
L-BFGS-B	1	90	0.673	39	0.011
L-BFGS-B	2	90	0.702	110	0.344
L-BFGS-B	3	90	0.696	128	0.789
COBYLA	1	90	0.691	128	0.522
COBYLA	2	90	0.712	128	0.978
COBYLA	3	90	0.718	128	1.000

respectively; the corresponding COBYLA values are 0.691, 0.712, and 0.718. On the 90 paired (instance, seed) comparisons per depth, COBYLA attains the larger approximation ratio in 50/90, 47/90, and 52/90 pairs for $p = 1, 2, 3$. These are descriptive paired counts rather than significance claims.

The cap is materially active. The fraction of runs ending at the evaluation ceiling for L-BFGS-B is 0.011, 0.344, and 0.789 across $p = 1, 2, 3$; for COBYLA it is 0.522, 0.978, and 1.000. Thus the ablation should be interpreted as attained quality under a finite optimization budget, not as a comparison of fully converged optimizers. In particular, the higher COBYLA medians in this budgeted experiment do not establish optimizer superiority outside this suite or budget.

Threshold-objective sensitivity. Under the matched 128-evaluation cap, direct event training raises median instance-level $P_{\tau_{0.40}}$ from 0.0683/0.1187/0.1007 to 0.0874/0.1618/0.1868 at $p = 1/2/3$. Median gains (+0.0159/+0.0173/+0.0174) all retain positive bootstrap intervals and survive Holm correction ($p = 0.0039/0.0020/0.0385$), while median approximation-ratio changes are -0.0012/-0.0045/-0.0140. Thus objective alignment improves the target event at a small expected-cut cost; the full 270-run paired ledger is retained in the package.

4.14. Instance-clustered inferential checks

The fixed-suite inferential analysis supports a narrower statement than the descriptive aggregate medians alone. For the primary L-BFGS-B depth study, the Friedman statistic across the 18 matched instance medians is $\chi_F^2 = 20.333$ with $p = 0.00004$. After Holm correction, all three pairwise depth contrasts remain below 0.05. The median paired approximation-ratio differences are +0.0234 for $p = 2$ versus $p = 1$, +0.0379 for $p = 3$ versus $p = 1$, and +0.0307 for

Table 16: Instance-clustered inferential checks for QAOA depth and optimizer family. Five optimizer starts are collapsed within each of the 18 fixed primary benchmark instances before testing. Confidence intervals resample instances; p values are Holm-adjusted within the three depth contrasts or the three optimizer contrasts.

Depth contrasts		95% CI	P_{Holm}	r_{rb}	n
contrast	median Δ				
p2-p1	+0.0234	[+0.0045,+0.0648]	0.0080	+0.731	18
p3-p1	+0.0379	[+0.0179,+0.1369]	0.0002	+0.942	18
p3-p2	+0.0307	[+0.0045,+0.0428]	0.0080	+0.743	18

Optimizer contrasts: COBYLA – L-BFGS-B					
p	median Δ	95% CI	P_{Holm}	r_{rb}	n
1	+0.0015	[-0.0000,+0.0281]	0.0911	+0.556	18
2	+0.0018	[-0.0001,+0.0086]	0.0911	+0.462	18
3	+0.0084	[-0.0007,+0.0175]	0.0911	+0.579	18

$p = 3$ versus $p = 2$. Their instance-bootstrap 95% intervals are [0.0045, 0.0648], [0.0179, 0.1369], and [0.0045, 0.0428], with Holm-adjusted p values 0.0080, 0.0002, and 0.0080, respectively. Thus the fixed suite supports ordered attained-quality differences across all three tested depths under this initialization protocol, without implying population-level monotonicity. The separate 20-start audit in Table 14 preserves the sign of every contrast and yields positive bootstrap intervals for all three differences, reducing concern that the conclusion is driven by the five-start initialization sample.

For the 128-evaluation optimizer ablation, the primary paired unit is also the instance: the five seed-level COBYLA-minus-L-BFGS-B differences are first reduced to an instance median. The median instance differences are +0.0015, +0.0018, and +0.0084 at $p = 1, 2, 3$. Their Holm-adjusted signed-rank p values are 0.0911, 0.0911, and 0.0911, respectively. All three 95% bootstrap intervals include zero (the $p = 1$ lower endpoint is numerically indistinguishable from zero). Consequently, this fixed 128-evaluation budget does not support a multiplicity-corrected optimizer-family difference at any tested depth, even though COBYLA has a positive descriptive median delta at each depth. Table 16 reports the paired effect sizes and intervals.

4.15. Classical references and interpretation boundary

The Goemans–Williamson layer provides a standard Max-Cut relaxation before the heuristic results are interpreted. All 18 primary and all 45 connected-validation SDP rows satisfy the declared 10^{-6} numerical primal–dual certificate. The maximum observed primal–dual gap is 2.3×10^{-7} in the primary tier and 4.4×10^{-7} in the connected-validation tier. The median SDP upper-bound ratio SDP/C^* is 1.015 and 1.017, respectively, showing that the relaxation is tight on these exact micro-instances while remaining a genuine upper bound. Across 4096 seeded random-hyperplane rounds per graph, the median mean rounded-cut ratio is 0.967 in the primary tier and 0.961 in the connected-validation tier. At least one rounded sample reaches the exact optimum on all 18 primary instances and 44/45 connected-validation instances. Table 17 summarizes this layer. The optimum-hit counts are

Table 17: Goemans–Williamson SDP and seeded random-hyperplane rounding. “cert.” counts rows with a numerical primal–dual gap no larger than 10^{-6} and dual-feasible slack. Exact C^* is used only for retrospective ratios.

Tier	Cert.	SDP/ C^*	Round/ C^*	Opt. hit
Primary	18/18	1.015	0.967	18/18
Connected	45/45	1.017	0.961	44/45

descriptive finite-sample results, not the Goemans–Williamson approximation guarantee and not evidence of scalable classical runtime.

The simple hill-climbing sanity reference is already strong on these micro-instances. Across all six instances for each size, its median local-search approximation ratio is 1.000 for $n = 8$, 1.000 for $n = 10$, and 0.977 for $n = 12$, with median optimum-hit fractions 0.613, 0.613, and 0.387. The objective-evaluation budget curve shows that the stronger heuristic conclusion is not an artifact of choosing 4096 evaluations. At only 64 complete objective evaluations per run, both simulated annealing and tabu search already have a median-of-instance-median run-best approximation ratio of 1.000; the corresponding median instance optimum-hit fractions are 0.555 and 0.789. Across 64 independent runs per instance, every one of the 18 primary instances is solved optimally at least once by both methods even at this smallest budget. At 128 evaluations, the median instance optimum-hit fraction is 0.688 for simulated annealing and 1.000 for tabu search; by 1024 evaluations it is 1.000 for both. The 4096 endpoint therefore confirms rather than creates the conclusion that these exact micro-instances are classically easy.

Figure 5 retains the original sanity comparison between the best QAOA depth by instance-level median and hill climbing. The machine-readable budget ledger retains the complete objective-evaluation curve rather than only its 4096 endpoint. Because evaluations can revisit states, the budget/state-space ratios are bookkeeping descriptors rather than exhaustive-search fractions. Consequently, the present data do not support a performance-advantage claim for QAOA or for the amplitude-amplification formulation. Instead, they show that the quantum-side experiments should be interpreted as configuration diagnostics rather than competitiveness evidence.

5. Discussion

5.1. What the resource model shows

The amplitude-amplification results show that a stopping rule based only on ideal rotation geometry is incomplete for a resource-aware workflow. More importantly, the operational construction separates the question “what target should be marked?” from knowledge of C^* . The oracle is defined by a cut-value comparator, while Eq. (6) fixes its threshold from graph-observable anchors. Exact enumeration is then used only to audit feasibility and marked fraction. The C^* -normalized q axis remains useful as a calibration experiment, but it is no longer conflated with deployable threshold construction.

The normalized-cost grid therefore remains a sensitivity statement rather than a processor model. The architecture-informed oracle layer sharpens this interpretation: once a con-

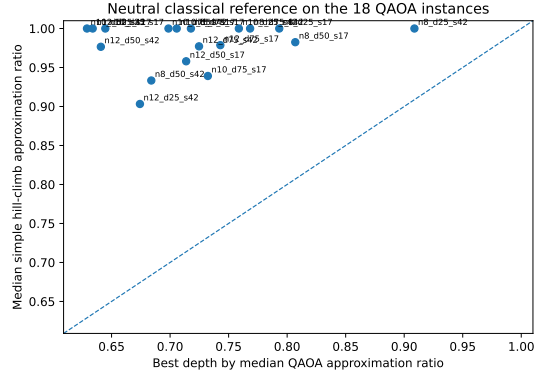


Figure 5: Neutral classical reference on the 18 QAOA instances. The horizontal coordinate is the best QAOA depth chosen by median approximation ratio for each instance; the vertical coordinate is the median approximation ratio of the simple multistart hill climber. The dashed line denotes equality.

crete ripple-style accumulator/comparator burden is charged and its logical depth is coupled to attenuation, the selector chooses no amplification in all 255 operational conditions when fixed per-trial work is set to zero. The fixed-overhead stress test shows why this boundary must be stated: the median zero-attenuation break-even fixed overhead is only 0.163 oracle-cost units in the primary tier, and nonzero amplification becomes common by $\chi = 0.25$. The result is therefore a conditional configuration map, not a claim that amplitude amplification is generally useless. The compiler-locked Qiskit layer independently confirms a large executable oracle burden under one fixed basis and synthetic topology, but it does not replace the logical selector or supply control/readout, device-duration, or calibration costs.

5.2. What the QAOA results show

The QAOA experiment reinforces the distinction between depth capacity and attained performance. Under the corrected instance-conditioned initialization stream, the aggregate median approximation ratio rises across all three tested depths, and all three fixed-suite paired depth contrasts survive Holm correction. This is a stronger within-suite pattern than in the earlier shared-initialization design, but it remains conditional on the 18 deterministic instances and does not establish population-level monotone QAOA behavior. The spread across seeded initializations still means that a single run is not a stable basis for comparison. The paired 128-evaluation ablation adds a second sensitivity axis: COBYLA has higher aggregate medians than L-BFGS-B at all three tested depths, yet none of the instance-clustered optimizer contrasts survives Holm correction and many deeper runs reach the cap. The result is therefore a finite-budget optimizer sensitivity analysis rather than an optimizer ranking. This is consistent with prior work on optimizer and initialization sensitivity [7, 8].

The benchmark therefore treats optimizer seed as an experimental factor, not an implementation detail. A hardware extension should retain the same discipline and additionally separate optimization shots, final evaluation shots, compiler seed, device calibration state, and queue/execution timing.

5.3. Why the classical reference matters

The classical references prevent a misleading interpretation of micro-instance quantum results. The certified SDP relaxation supplies a canonical Max-Cut upper bound and its hyperplane rounding produces high-quality feasible cuts without using C^* ; the simple hill climber is frequently exact, and both fixed-budget simulated annealing and tabu search have median run-best approximation ratio 1.000 with median optimum-hit fraction 1.000 at every tested size. This finding is not a failure of the benchmark; it is precisely why neutral baselines are necessary. A quantum method should not be described as advantageous simply because it improves over uniform sampling or because its internal hit probability increases when multiple transparent classical heuristics already solve the same small instances reliably.

The current study therefore separates two questions that are often conflated. The first is a configuration question: how do quantum algorithm parameters respond to target quality and resource assumptions? The second is a comparative-performance question: does the resulting workflow outperform suitable classical alternatives under matched physical resources? The common P_τ metric aligns the success event but does not solve the second problem because physical costs remain unmatched. The present results therefore strengthen fairness of interpretation without converting the study into a performance-advantage claim.

5.4. Relevance to future-generation quantum systems

The systems relevance lies in the explicit interface between algorithmic parameters and resource assumptions. Hybrid quantum platforms must eventually map abstract oracle calls, variational layers, classical control, and scheduling into concrete execution resources. Recent FGCS work emphasizes precisely this quantum–classical integration, simulation, and infrastructure perspective [4, 3, 5]. The current benchmark provides a small, fully auditable layer beneath such system studies: exact problem ground truth, reproducible parameter sweeps, a separate seed/topology coverage stress test, and a claim policy that prevents model variables from being relabeled as empirical hardware measurements. The additional coverage tier is deliberately kept separate from the 18-instance primary QAOA inference so that broader descriptive stress testing does not retroactively inflate the inferential sample.

6. Threats to validity and limitations

The exact-comparison tier is intentionally small in vertex count. Exact primary ground truth remains restricted to at most 12 vertices, while the connected non-bipartite validation tier extends exact structural validation only to $n = 14$; no Max-Cut scaling claim follows from either tier. The separate $n = 14$ –20 host-simulation diagnostic measures classical reference cost only, and the $n_{\text{model}} = 12$ –24 analytical tier uses abstract marked fractions rather than Max-Cut optima. These tiers expand the evidence boundary without pretending that extrapolative model rows are exact optimization results.

The structural audit also shows that the original primary and expanded coverage designs include disconnected and bipartite instances. Those rows are retained for traceability rather than removed after inspection, and the 45-instance connected non-bipartite tier is reported separately as a sensitivity control. Conditioning on those two properties does not establish instance hardness and does not make the validation tier a probability sample over graph populations. The dense operational- λ audit reduces dependence on the three displayed operating points, but it does not turn λ into a universally optimal application parameter; choosing a target strictness remains workload- and utility-dependent. Seed/topology coverage is broader than in the primary design—five random seeds and three structured families produce 63 exact coverage instances—but it is still a deliberately finite, hand-declared benchmark. The structured families use only two weight seeds, and no claim is made that cycle, 3-regular circulant, and two-community graphs exhaust relevant Max-Cut topology classes.

The amplitude-amplification oracle now marks a numeric threshold event $C(x) \geq \tau$ rather than a pre-enumerated optimum set, and the operational threshold in Eq. (6) does not use C^* . The BBHT layer removes exact ρ_τ from the executed growth schedule, while exact ρ_τ remains necessary only for retrospective expected-cost scoring in the present exact tiers. This is an algorithmic unknown-solution-count control, not a hardware implementation of random quantum-control, measurement, verification, or adaptive host–device latency. Quantum counting and fixed-point search are not implemented. The reversible weighted-cut accumulator/comparator template remains architecture-informed bookkeeping rather than compiler output; the separate Qiskit layer uses an independently executable arithmetic/comparator realization and is not used to retrofit the analytical count formulas. The factor used to form G_{eq} is a transparent weighting convention, not a physical native-gate equivalence. The $\alpha_T \in \{1, 4, 6, 10\}$ sensitivity sweep shows that the zero-fixed-overhead $k^* = 0$ boundary is invariant over this range, but the fixed-overhead break-even coordinate moves materially; therefore no single scalarization is presented as physically privileged. The compiler control fixes one basis, one seed, and a synthetic line topology, but alternative synthesis, connectivity, fault-tolerant realization, and backend timing can materially change the concrete counts. The attenuation variables are synthetic; neither the abstract $D_{\text{prep}}, D_{\text{iter}}$ sensitivity coordinates nor the architecture-coupled ε_D are measured device quantities. The fixed-overhead coordinate $\chi = C_{\text{fixed}}/G_O$ is likewise synthetic: it demonstrates sensitivity to omitted per-trial work but does not estimate control, measurement, verification, network, queue, or host-device handoff cost on any platform.

The QAOA experiments use exact statevectors. The primary depth study uses L-BFGS-B with a five-start canonical analysis and a nested 20-start stability audit, while the optimizer ablation compares L-BFGS-B with COBYLA under one finite cap of 128 exact expectation evaluations and five paired initializations. The statistical layer collapses repeated starts within each of the 18 primary instances and therefore avoids treating optimizer seeds as independent graph replicates;

nevertheless, 18 fixed instances still provide limited inferential resolution and do not support population-level generalization. This broadens the optimizer evidence but still does not characterize the full optimization landscape, and the cap binds many $p = 2$ and $p = 3$ runs. Hardware noise, finite-shot optimization, parameter-transfer methods, alternative mixers, advanced initialization strategies, and additional optimizer families remain outside the current evidence set. The classical layer now contains a numerically certified Goemans–Williamson SDP/rounding reference together with hill climbing and a six-point simulated-annealing/tabular objective-evaluation budget curve, but it is still not a comprehensive classical benchmark: no branch-and-bound timing study, industrial Max-Cut solver comparison, or large-instance heuristic competition is claimed. The classical budget counts complete objective evaluations rather than unique visited states, and it is deliberately not mapped to QAOA objective calls, quantum shots, or amplitude-amplification oracle resources. The SDP implementation is numerically certified to a declared tolerance rather than by symbolic exact arithmetic, and its 4096 hyperplane rounds are a finite randomized sample. Exact enumeration supplies micro-instance ground truth and is not presented as a scalable competitor.

Finally, the benchmark does not compare physical energy consumption, monetary cost, or elapsed quantum-service time. Those quantities require a different evidence layer with explicit backend and system telemetry.

7. Conclusion

This paper presented a reproducible benchmark for threshold-oriented resource-aware amplitude amplification and QAOA with explicitly separated exact, host-simulation, and analytical tiers. The main threshold-construction result is operational rather than optimum-normalized: τ_λ^{op} is built from the uniform-random expected cut and a Laplacian spectral upper bound without using C^* . Retrospective exact validation shows that $\lambda = 0.25$ and 0.40 are feasible on all 18 primary benchmark instances, whereas $\lambda = 0.55$ is feasible on 15/18, making target infeasibility visible instead of silently assuming attainability. A separate 21-point audit from $\lambda = 0$ to 1 in steps of 0.05 shows that feasibility decreases beyond these declared points rather than appearing only at the selected levels; at $\lambda = 0.75$ feasibility is 5/18 primary and 12/45 connected-validation instances, and at $\lambda = 1$ it is 2/18 and 0/45. A separate exact coverage stress test broadens the seed set and adds cycle, 3-regular circulant, and two-community topologies: the first operational level is feasible on all 63 coverage instances, the second on 62/63, the strictest level on 55/63, and all 360 architecture-coupled feasible coverage conditions at $\varepsilon_D \in \{0, 10^{-3}\}$ retain $k^* = 0$ under the zero-fixed-overhead convention. Because a structural audit finds five disconnected and two bipartite graphs in the primary tier, a separate 45-instance connected, non-bipartite exact validation tier through $n = 14$ was added without replacing the original benchmark; all 252 feasible architecture-coupled conditions in that tier also retain $k^* = 0$ at zero fixed overhead. A separate 8,085-condition

fixed-overhead stress test shows that the zero-round conclusion is conditional: at zero attenuation, median break-even overhead ratios are 0.163 and 0.168 oracle-cost units for the primary and coverage tiers. A separate C^* -normalized calibration study contains 864 coarse and 11,934 dense sensitivity conditions and confirms that target strictness, synthetic attenuation, and normalized oracle cost materially change the selected amplification depth. Across 18 statevector QAOA instances and 270 primary L-BFGS-B optimization runs, aggregate median approximation ratios are 0.673/0.689/0.716 at $p = 1/2/3$. A nested 1,080-run, 20-start audit gives 0.661/0.691/0.707 and retains all three positive instance-clustered depth contrasts after Holm correction, with all bootstrap intervals above zero. A separate 540-run paired optimizer ablation under a 128-objective-evaluation cap yields aggregate medians of 0.673/0.702/0.696 for L-BFGS-B and 0.691/0.712/0.718 for COBYLA, while also showing substantial cap exhaustion at the deeper settings. Instance-clustered inference further narrows the interpretation: the depth omnibus test is significant on the fixed suite ($p = 0.00004$), and all three pairwise depth contrasts survive Holm correction. Under the finite optimizer budget, none of the COBYLA-minus-L-BFGS-B contrasts survives Holm correction; this remains a budget-conditioned attained-quality comparison rather than evidence for a general optimizer ranking. The classical conclusion remains strong: the Goemans–Williamson SDP is numerically certified on all 63 primary/connected-validation rows, with median SDP-to-exact-optimum ratios of 1.015 and 1.017 and an exact optimum found by finite hyperplane rounding on 62/63 instances. More importantly, the heuristic conclusion survives a six-point objective-evaluation budget curve: at only 64 evaluations per run, simulated annealing and tabu search both have a median-of-instance-median approximation ratio of 1.000 and solve all 18 primary instances at least once across 64 runs, with median instance optimum-hit fractions of 0.555 and 0.789. These results reinforce that the exact tiers are configuration tests rather than evidence for quantum competitiveness.

The primary contribution is therefore a benchmark discipline rather than a speedup claim: construct operational thresholds without C^* , distinguish threshold construction from exact-tier validation, quantify fixed-depth regret under marked-fraction misspecification, evaluate an information-free BBHT stopping policy, align methods on a common target-hit event, and then test whether conclusions survive an explicit reversible-oracle burden. In the architecture-informed logical layer developed here, the oracle median grows from 420 to 1,088 modeled Toffoli gates across $n = 8$ to 12 , and all 255 coupled operational conditions select the no-amplification configuration when fixed per-trial overhead is zero. The fixed-overhead phase boundary is retained alongside this negative result because it shows exactly where that configuration conclusion changes rather than treating one bookkeeping convention as universal. A separate Toffoli-weight sweep confirms that the zero-overhead boundary survives $\alpha_T = 1$ – 10 while the break-even location shifts, making the scalarization dependence explicit rather than hidden. The added Qiskit 2.4.2 evidence layer independently transpiles 54 routed QAOA circuits and

12 threshold-oracle cases; its much larger executable oracle depths/CX counts corroborate the qualitative burden while remaining explicitly separate from hardware performance.

Data and code availability

All primary, expanded-coverage, and connected non-bipartite validation benchmark instances, machine-readable edge ledgers, structural graph-diagnostic ledgers, exact ground-truth records, operational and benchmark-normalized threshold definitions, retrospective operational-feasibility records, marked-fraction misspecification rows, BBHT unknown-solution-count expected-cost rows, logical resource-scalarization sensitivity rows, dense operational-lambda audit rows, common target-metric rows, raw QAOA runs, the 20-start initialization-stability ledger and comparison, paired optimizer-budget runs and summaries, instance-clustered inferential ledgers, raw and summarized classical objective-evaluation budget-curve runs, analytical sensitivity outputs, compiler-locked Qiskit ledgers and environment records, the executable compiler-validation script/workflow, generated tables, test results, the exact environment lock, the automated preflight validator, and a hash-bearing run manifest are included in the accompanying reproducibility package.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

Ercan Erkalkan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing, Project administration.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT for manuscript organization, drafting and language refinement, code drafting, test design, literature-navigation support, and consistency checking. The final reviewer-driven audit and revision on 1 September 2026 used ChatGPT with the GPT-5.6 Sol model; exact model-version identifiers for earlier ChatGPT-assisted sessions were not consistently recorded and are therefore not inferred retrospectively. After using this tool, the author reviewed and edited the content as needed, executed and verified the computational workflow and reported outputs, checked the cited sources, and takes full responsibility for the content of the publication.

References

- [1] Lov K. Grover, A fast quantum mechanical algorithm for database search, in: *Proceedings of the Twenty-Eighth Annual ACM Symposium on Theory of Computing*, 1996, pp. 212–219. doi:10.1145/237814.237866.
- [2] Gilles Brassard, Peter Høyer, Michele Mosca, Alain Tapp, Quantum Amplitude Amplification and Estimation, in: *Quantum Computation and Information*, Vol. 305 of *Contemporary Mathematics*, American Mathematical Society, 2002, pp. 53–74. doi:10.1090/conm/305/05215.
- [3] Gilbert Netzer, Pratibha Raghupati Hegde, Ivy Peng, Stefano Markidis, Tightly-integrated quantum–classical computing using the QHDL hardware description language, *Future Generation Computer Systems* 174 (2026) 107977. doi:10.1016/j.future.2025.107977.
- [4] Stefano Markidis, Lucio Grandinetti, Michela Taufer, Editorial on Future Generation Computer Systems special collection on advances in quantum computing: methods, algorithms, and systems Vol II, *Future Generation Computer Systems* 174 (2026) 107993. doi:10.1016/j.future.2025.107993.
- [5] Marzio Vallero, Paolo Rech, Flavio Vella, State of practice: Evaluating GPU performance of state vector and tensor network methods, *Future Generation Computer Systems* 174 (2026) 107927. doi:10.1016/j.future.2025.107927.
- [6] Edward Farhi, Jeffrey Goldstone, Sam Gutmann, A Quantum Approximate Optimization Algorithm, *arXiv preprint arXiv:1411.4028* (2014).
- [7] Mario Fernández-Pendás, Elías F. Combarro, Sofia Vallecorsa, José Ranilla, Ignacio F. Rúa, A study of the performance of classical minimizers in the Quantum Approximate Optimization Algorithm, *Journal of Computational and Applied Mathematics* 404 (2022) 113388. doi:10.1016/j.cam.2021.113388.
- [8] Stefan H. Sack, Maksym Serbyn, Quantum annealing initialization of the quantum approximate optimization algorithm, *Quantum* 5 (2021) 491. doi:10.22331/q-2021-07-01-491.
- [9] Yu Pan, Yifan Tong, Yi Yang, Automatic depth optimization for a quantum approximate optimization algorithm, *Physical Review A* 105 (2022) 032433. doi:10.1103/PhysRevA.105.032433.
- [10] Theodore J. Yoder, Guang Hao Low, Isaac L. Chuang, Fixed-Point Quantum Search with an Optimal Number of Queries, *Physical Review Letters* 113 (2014) 210501. doi:10.1103/PhysRevLett.113.210501.
- [11] Michel Boyer, Gilles Brassard, Peter Høyer, Alain Tapp, Tight Bounds on Quantum Searching,

- Fortschritte der Physik 46(4–5) (1998) 493–505. doi:10.1002/(SICI)1521-3978(199806)46:4/5<493::AID-PROP493>3.0.CO;2-P.
- [12] Kostas Blekos, Dean Brand, Andrea Ceschini, Chiao-Hui Chou, Rui-Hao Li, Komal Pandya, Alessandro Summer, A review on Quantum Approximate Optimization Algorithm and its variants, *Physics Reports* 1068 (2024) 1–66. doi:10.1016/j.physrep.2024.03.002.
 - [13] Lixue Cheng, Yu-Qin Chen, Shi-Xin Zhang, Shengyu Zhang, et al., Quantum approximate optimization via learning-based adaptive optimization, *Communications Physics* 7 (2024). doi:10.1038/s42005-024-01577-x.
 - [14] Vivek Katial, Kate Smith-Miles, Charles Hill, Lloyd Hollenberg, On the Instance Dependence of Parameter Initialization for the Quantum Approximate Optimization Algorithm: Insights via Instance Space Analysis, *INFORMS Journal on Computing* (2024). doi:10.1287/ijoc.2024.0564.
 - [15] Danylo Lykov, Jonathan Wurtz, Cody Poole, Mark Saffman, Tom Noel, Yuri Alexeev, Sampling frequency thresholds for the quantum advantage of the quantum approximate optimization algorithm, *npj Quantum Information* 9 (2023) 73. doi:10.1038/s41534-023-00718-4.
 - [16] Sandeep Suresh Cranganore, Vincenzo De Maio, Ivona Brandic, Ewa Deelman, Paving the way to hybrid quantum–classical scientific workflows, *Future Generation Computer Systems* 158 (2024) 346–366. doi:10.1016/j.future.2024.04.030.
 - [17] Thomas L. Beck, Alessandro Baroni, Ryan Benink, Gilles Buchs, et al., Integrating quantum computing resources into scientific HPC ecosystems, *Future Generation Computer Systems* 161 (2024) 11–25. doi:10.1016/j.future.2024.06.058.
 - [18] Vlad Gheorghiu, Michele Mosca, Quantum resource estimation for large scale quantum algorithms, *Future Generation Computer Systems* 162 (2025) 107480. doi:10.1016/j.future.2024.107480.
 - [19] José A. Tirado-Domínguez, Eladio Gutiérrez, Oscar Plata, QTIS: A QAOA-based Quantum Time Interval Scheduler, *Future Generation Computer Systems* 183 (2026) 108594. doi:10.1016/j.future.2026.108594.
 - [20] Timothy Proctor, Kevin Young, Andrew D. Baczewski, Robin Blume-Kohout, Benchmarking quantum computers, *Nature Reviews Physics* 7 (2025) 105–118. doi:10.1038/s42254-024-00796-z.
 - [21] Ang Li, Samuel Stein, Sriram Krishnamoorthy, James Ang, QASMBench: A Low-Level Quantum Benchmark Suite for NISQ Evaluation and Simulation, *ACM Transactions on Quantum Computing* 4(2) (2023) 1–26. doi:10.1145/3550488.
 - [22] Teague Tomesh, Pranav Gokhale, Victory Omole, Gokul Subramanian Ravi, Kaitlin N. Smith, Joshua Vizslai, Xin-Chuan Wu, Nikos Hardavellas, Margaret Martonosi, Frederic T. Chong, SupermarQ: A Scalable Quantum Benchmark Suite, in: *2022 IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, 2022, pp. 587–603. doi:10.1109/HPCA53966.2022.00050.
 - [23] Thomas Lubinski, Sonika Johri, Paul Varosy, Jeremiah Coleman, Luning Zhao, Jason Necaie, Charles H. Baldwin, Karl Mayer, Timothy Proctor, Application-Oriented Performance Benchmarks for Quantum Computing, *IEEE Transactions on Quantum Engineering* 4 (2023) 1–32. doi:10.1109/TQE.2023.3253761.
 - [24] Thomas Lubinski, Joshua J. Goings, Karl Mayer, Sonika Johri, Nithin Reddy, Aman Mehta, Niranjana Bhatia, Sonny Rappaport, Daniel Mills, Charles H. Baldwin, Luning Zhao, Aaron Barbosa, Smarak Maity, Pranav S. Mundada, Quantum Algorithm Exploration using Application-Oriented Performance Benchmarks, *arXiv preprint arXiv:2402.08985* (2024). doi:10.48550/arXiv.2402.08985.
 - [25] Steven A. Cuccaro, Thomas G. Draper, Samuel A. Kutin, David Petrie Moulton, A new quantum ripple-carry addition circuit, *arXiv preprint quant-ph/0410184* (2004). doi:10.48550/arXiv.quant-ph/0410184.
 - [26] Maxime Remaud, Optimizing T and CNOT Gates in Quantum Ripple-Carry Adders and Comparators, in: *Proceedings of the 2024 Workshop on Recent Advances in Quantum Computing and Technology*, 2024. doi:10.1145/3665870.3665875.
 - [27] Michel X. Goemans, David P. Williamson, Improved approximation algorithms for maximum cut and satisfiability problems using semidefinite programming, *Journal of the ACM* 42(6) (1995) 1115–1145. doi:10.1145/227683.227684.